James LaFemina

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Education

Ph.D. Student, Dept. of Biomolecular Engineering
University of California, Santa Cruz 2023 – Present
Advisor: Professor Tal Sharf

B.S. Biochemistry & Molecular Biology, Highest Honors in the Major
University of California, Santa Cruz (2023)

Research Experience

Sharf Lab, Dept. of Biomolecular Engineering, UC Santa Cruz

Interdisciplinary mission to decode how human brain circuits emerge, interact, and transmit informa8on by combining 3D stem cell–derived brain models with high-resolu8on neuro-electronics to map ac8vity at single-cell resolu8on across func8onal networks.

Graduate Researcher | Sept 2024 – Present

My research aims to uncover the early drivers of Parkinson's disease (PD) using genomics and systems approaches to link molecular mechanisms, gene regulatory dynamics, and circuit-level ac>vity in human-derived brain models—ul>mately enabling discovery of disease-modifying interven>ons.

Selected Achievements:

- Op>mized midbrain organoid differen>a>on protocols (including neurotoxic lesioning) to develop late-stage PD
  in vitro models.
- Developed and executed immunohistochemistry workflows to visualize and quan>fy dopaminergic neuron deple>on in organoids.
- Performed high-density microelectrode array (HD-MEA) experiments on fused midbrain-thalamic organoids (assembloids).
- Constructed a custom data analysis pipeline using Python programming and sta>s>cal methods to characterize func>onal network breakdown following neurotoxin treatment.
- Collaborated with fabrica>on, electrophysiology, and computa>onal teams within the Braingeneers consor>um to itera>vely refine experimental design, data-analysis approaches, and interpreta>on of results.

Boeger Lab, Dept. of Molecular, Cell, and Developmental Biology, UC Santa Cruz

Research aimed at understanding how stochas8c transcrip8onal bursts and ATP-dependent chroma8n remodeling together shape gene regula8on specificity, using a combina8on of theore8cal modeling and single-molecule analyses to reveal how dynamic chroma8n states resolve molecular noise into precise cellular outcomes.

Graduate Researcher | June 2023 – Sept 2024

I inves>gated how disrup>ons in histone acetyla>on and nucleosome dynamics destabilize chroma>n architecture and drive aneuploidy in budding yeast. To dissect this rela>onship, I used reverse gene>cs, chroma>n biochemistry, quan>ta>ve single-cell analysis, and sta>s>cal modeling to link molecular modifica>ons at the nucleosome level to large-scale genomic instability.

Selected Achievements:

- Designed experiments to test whether improper H3K56 acetyla>on induces re-replica>on rather than chromosomal missegrega>on as the primary driver of aneuploidy.
- Developed mutant yeast strains mimicking cons>tu>ve H3K56 hypo- and hyperacetyla>on to test their impact on genome stability.
- Employed a fluorescent reporter system to infer chromosomal copy number varia>ons via flow cytometry, enabling real->me tracking of aneuploidy in single cells.
- Validated this mechanism by adap>ng and modernizing the Luria-Delbrück fluctua>on test to quan>fy muta>on rates under different acetyla>on states.
- Applied topological assays using DNA supercoiling and topoisomerases to quan>fy nucleosome occupancy under various gene>c perturba>ons.
- Used MNase diges>on and nanoNOMe (single-molecule chroma>n accessibility and methyla>on sequencing) to analyze chroma>n structure and accessibility.
- Engineered inducible anchor-away systems to acutely deplete key regulatory proteins (e.g., TBP) and assess nucleosome dynamics in real >me

Undergraduate Researcher | March 2023 – June 2023

Senior Thesis: *ASF1-Mediated Histone H3K56 Regula8on in Yeast Chroma8n Stability*

Advisor: Professor Hinrich Boeger

Teaching and Mentorships

Teaching Assistant

University of California, Santa Cruz

- *Biochemistry III* (Spring 2023 and Spring 2024)
- *Eukaryo2c Molecular Biology* (Winter 2024)

Undergraduate Mentorships

- Mentored a total of 6 undergraduates at my bench in the Boeger Lab.

Professional Summary (Prior Career)

Sales executive and leadership development specialist with a 20-year track record of helping senior leaders at some of the world's top brands solve complex problems with business intelligence, software and analytics tools, and experiential learning programs.

Selected Achievements:

- Led enterprise sales of high-impact experien>al leadership programs in partnership with NASA targe>ng senior leadership teams and high-poten>al cohorts at Fortune 500s.
- Collaborated directly with execu>ves to shape learning engagements that addressed pressing organiza>onal challenges and promoted adap>ve team leadership..
- Built and managed pipelines as both an internal consultant and external on-demand VP, helping startups and Fortune 500 clients alike solve high-stakes leadership and performance challenges.